DO NOW — RECALL

Floating point, binary and Boolean mathematics

Name: _____ Class: _____ Date: _____

1. What is normalisation's purpose?

2. What is first-two-bit rule for normalised two's-complement mantissa?

3. State De Morgan: NOT(A AND B).

4. Simplify (A∧B)∨(¬A∧B) with working.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Standardised mantissa maximising precision.

2. Model answer

The first two bits differ.

3. Model answer

NOT A OR NOT B.

4. Model answer

$B \wedge (A \vee \neg A) = B$.

By the end, you can…

• Normalising signed binary mantissas shifts bits and compensates exponent to preserve the same value.

• For negative integer conversion, write exactly eight positive bits, invert each, then add one.

• For algebraic simplification, show each Boolean law used rather than only the final expression.

Vocabulary — know these terms

mantissa — Significand representing precision.

exponent — Power-of-two scale.

two's complement — Representation for signed integers.

normalisation — Shift/scaling preserving represented value.

NEW LEARNING

Read, annotate and explain

Understand

Normalising signed binary mantissas shifts bits and compensates exponent to preserve the same value.

Apply

For negative integer conversion, write exactly eight positive bits, invert each, then add one.

Evaluate

For algebraic simplification, show each Boolean law used rather than only the final expression.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

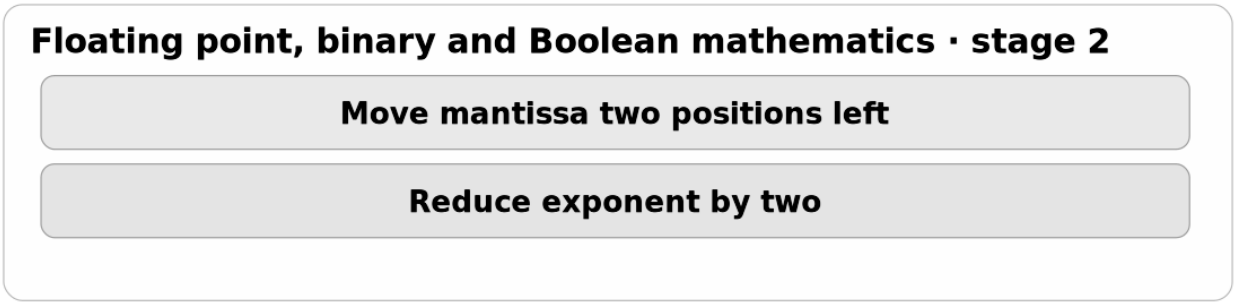
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



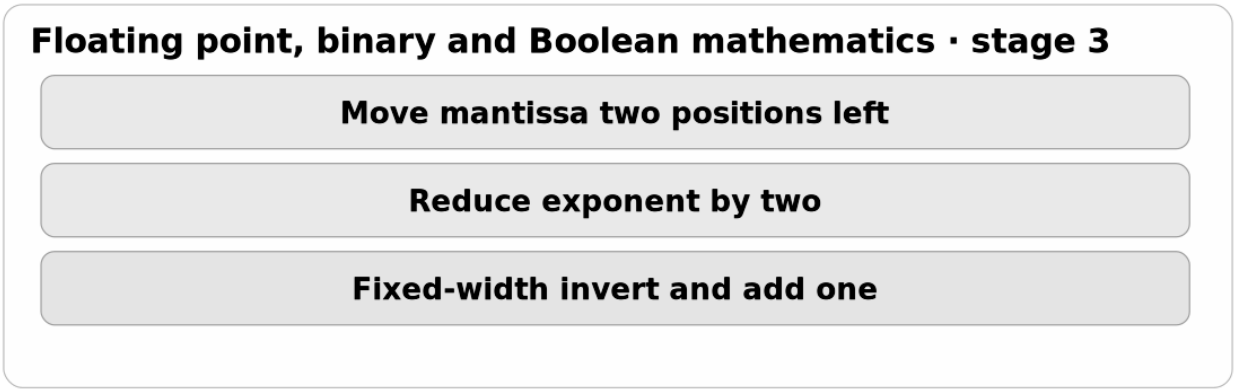
Floating point, binary and Boolean mathematics · stage 1: Move mantissa two positions left

New Learning



Floating point, binary and Boolean mathematics · stage 2: Move mantissa two positions left; Reduce exponent by two

New Learning

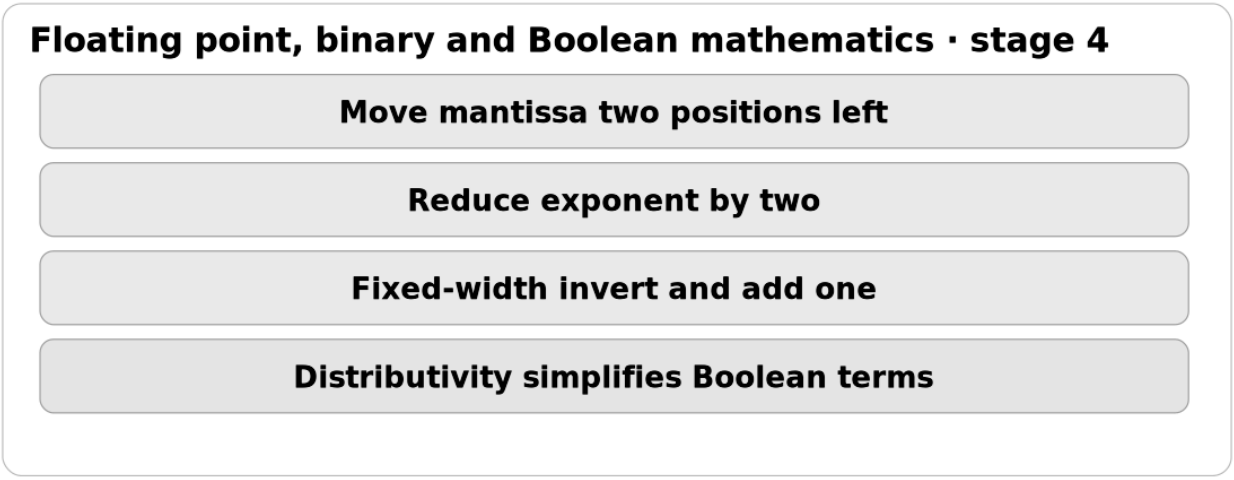


Floating point, binary and Boolean mathematics · stage 3: Move mantissa two positions left; Reduce exponent by two; Fixed-width invert and add one

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Floating point, binary and Boolean mathematics · stage 4: Move mantissa two positions left; Reduce exponent by two; Fixed-width invert and add one; Distributivity simplifies Boolean terms

I DO — WORKED EXAMPLE

Follow the worked method

Normalise $0.0101_2 \times 2^4$.

Predict first

Left-shift mantissa once and lower exponent by one.

Explain and check

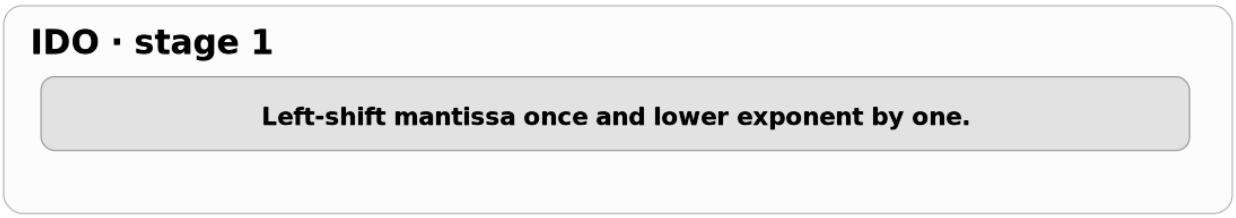
Explain the reason for the result.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Normalise $0.0101_2 \times 2^4$.

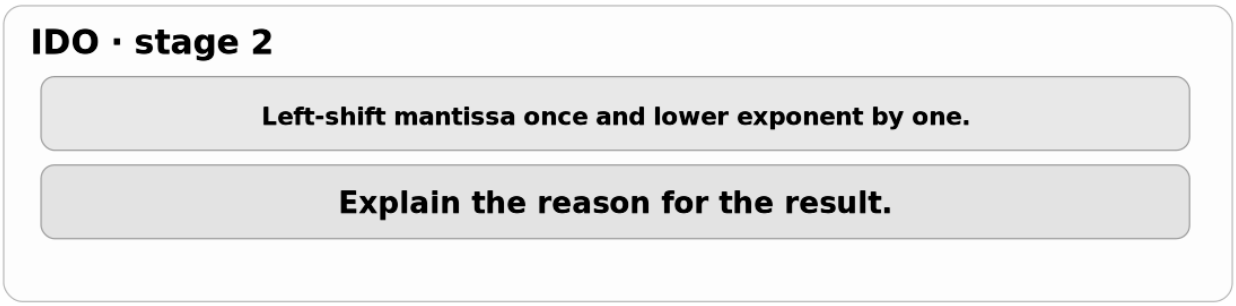
Left-shift mantissa once and lower exponent by one.



IDO · stage 1: Left-shift mantissa once and lower exponent by one.

Normalise $0.0101_2 \times 2^4$.

Left-shift mantissa once and lower exponent by one.



IDO · stage 2: Left-shift mantissa once and lower exponent by one.; Explain the reason for the result.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find 8-bit two's complement of −11.

Invert all eight bits then add one, yielding 11110101.



WEDO · stage 1: Invert all eight bits then add one, yielding 11110101.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Invert all eight bits then add one, yielding 11110101.

DISCUSS — Compare intermediate values and your reasoning. Justify one step using the relevant rule, representation or truth table.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Normalise $0.00101_2 \times 2^5$ and express its decimal value. [6]

Question 2

Convert −23 to 8-bit two's complement, showing every step. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Simplify A∨(¬A∧B) using identities; verify for A0,B1. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Normalisation shifts mantissa but never changes exponent.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find 8-bit two's complement of −11.

Invert all eight bits then add one, yielding 11110101.



WEDO · stage 2: Invert all eight bits then add one, yielding 11110101.; Explain the reason for the result.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain and check

Explain the reason for the result.

IMPROVE: Explain one correction to your own reasoning.